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Final project

An end-to-end machine learning study on a dataset you have not seen in class, written up so that another student could reproduce it.

Presented at **Lesson 1** so you can think about it while the course runs. Topic confirmed by **Lesson 4**. Submitted after Lesson 10.

What you have to do

Take one of the datasets below, or propose your own, and carry out a complete study:

1. **Frame the problem.** What is predicted, from what, and why would anyone want it? State the target, the features, and what a useful result would look like.
2. **Explore and prepare the data.** Missing values, outliers, encoding, scaling. Document every decision and its justification.
3. **Establish a baseline.** The simplest thing that could work — the majority class, the mean, a single feature. Everything later is measured against this.
4. **Design the evaluation before you model.** Choose your split or cross-validation scheme, and your metric, and justify both *given the problem*. Write this down before you fit anything serious.
5. **Compare at least three model families** from the course, tuned honestly.
6. **Diagnose.** Learning curves, error analysis, where and why the best model fails.
7. **Interpret.** What drives the predictions? What would you tell someone who had to act on this?
8. **State the limitations.** What would you need to trust this in production? What could go wrong, and who would it affect?

Deliberately not required: a state-of-the-art result. A careful study that concludes “this dataset does not support a useful model, and here is the evidence” is a complete and successful project.

Datasets

Pick one. All are public, load in seconds, and are small enough to run on a laptop.

Dataset	Task	Why it is interesting	Source
Bike sharing demand	Regression	Strong temporal structure — a trap for naive random splitting	UCI, id 275
Adult census income	Binary classification	Class imbalance plus genuine fairness questions across protected attributes	UCI, id 2
Online shoppers intention	Binary classification	Heavily imbalanced; accuracy is actively misleading here	UCI, id 468
Wine quality	Regression or ordinal	Small, correlated features, ambiguous target definition	UCI, id 186
Steel plates faults	Multiclass	Seven classes, unevenly represented, industrial context	UCI, id 198

Load any of them with:

```
from sklearn.datasets import fetch_openml

data = fetch_openml(data_id=..., as_frame=True)
X, y = data.data, data.target
```

Proposing your own dataset is welcome and must be agreed by Lesson 4. It has to be public, licensed for teaching use, and non-trivial — no Titanic, no Iris, no MNIST.

What to hand in

A folder named {surname}_{studentid} containing:

File	Contents
report.ipynb	The study: narrative and code together, running top to bottom in the course container
README.md	Half a page: the question, what you did, what you found, how to run it

Constraints:

- **Runs in the course Docker image with no extra installs.** If it needs a package we do not ship, it does not run.
- **Reproducible:** seeds fixed, and a second execution gives the same numbers.
- Outputs left in place, so it can be read without being executed.
- Roughly 2,000-3,000 words of narrative. Prose that explains decisions, not comments restating the code.

Assessment criteria

Criterion	Weight	What earns marks
Methodological soundness	35%	Evaluation designed before modelling; no leakage; the split scheme suits the data; honest treatment of uncertainty
Technical execution	25%	Correct implementation, sensible preprocessing, fair comparison between models
Diagnosis and interpretation	20%	Understands <i>why</i> the model behaves as it does; error analysis goes beyond a single metric
Communication	15%	A reader can follow the reasoning and reproduce the work
Critical judgement	5%	Limitations stated honestly, including what would make the result untrustworthy

Note what is absent: **there are no marks for accuracy**. Two students on the same dataset with different scores can both receive full marks, and the higher score can be the lower mark if it was obtained by contaminating the evaluation.

Automatic deductions

Problem	Effect
Test set used for any fitting decision — scaling, feature selection, tuning	Methodology capped at 40%
Results not reproducible on re-run	Execution capped at 50%
Notebook does not run in the course image	Returned unmarked for resubmission

Peer review

Before submission, each project is read by two other students, and each student reads two projects. Use [peer_review_form.md](#).

This is part of the assessment, not a formality: reviewing someone else's evaluation scheme is one of the fastest ways to learn to see the flaws in your own. The reviews you write are assessed alongside the project you submit.